MALIS Project 1: Reinforcement Learning and Multi-Armed Bandits

In this year's project, we will study reinforcement learning. To that end, in this first project, you will focus on the multi-armed bandit problem, starting from a game you play yourself.

To have you started a Python simulator of the game you will play (fishing.py) and the skeleton files that you will complete (agents.py, experiment.ipynb) are available on Moodle. You must use the skeleton files.

Please make sure that you register in a group by October 21st.

Objectives

By executing this project, you will be able to:

1. Understand the trade-off between exploration and exploitation in sequential decision-making.
2. Formalize a decision strategy as an algorithm.
3. Implement and compare bandit algorithms.
4. Design experiments that separate the effect of decisions from the effect of chance.
5. (Optional) Combine supervised learning with exploration in a contextual bandit.

Part I: Playing the game

Task 1. Each team member plays Fishing Season on Moodle once, **before 21 Oct 2026 at 23:59**, before reading Part III. On the final screen of the game, click "Download my log" to save your log as a JSON file. You can download it again later by reopening the activity on Moodle.

Part II: Human play

Use your team logs.

Task 2. Describe how each team member played each season and compare with their strategy notes. Abstract the strategies used in your team into precise rules that could be applied to any log.

Task 3. Propose a performance measure that separates decisions from luck. Use the simulator (see fishing.py) to show that it does, for example by playing the same strategy on many random bays. Compare your team's behavior across the three seasons: the effect of the shorter Season 2, and the change in Season 3 (what it is, which team members adapted, how fast).

Part III: Bandit algorithms

Task 4. Read Chapter 2 (Sections 2.1 to 2.7) of *Reinforcement Learning: An Introduction*, R. S. Sutton and A. G. Barto, 2nd edition (link).

In the skeleton file agents.py, implement as agents the best human strategy from Part II and at least two methods from the reading. The skeleton defines how an agent interacts with the simulator: follow it and do not modify its provided parts. Document your code.

Task 5. Using the skeleton notebook experiment.ipynb and the agents from agents.py, evaluate every agent on at least 1000 random bays per season and report the distribution of results. Then compare your agents with your team members on the bays they played, to see whether and where they do better. From these results, identify the best agent for Season 2 and for Season 3. Finally, choose one method and study how its performance depends on one of its parameters, over a range of values and for each season.

Failing to use agents.py will result in a penalty (i.e., no recoding in the notebook).

Optional - Part IV: Contextual bandits (required for a grade above 16/20)

The simulator includes a contextual version (you need to inspect the code). Each morning, before you choose a spot, it provides the wind speed, the wave height, and the tide level. The average catch at each spot depends on these conditions, and it varies by spot. Seasons in this version last 300 days.

Task 6. Predict the catch at each spot from the observed conditions with linear regression or KNN, and use the predictions to choose a spot, with agents written in agents.py. Compare with the best agent of Part III. Show a case where acting only on the predictions fails, and propose a fix.

Report:

You need to prepare a report of 1500 words maximum (figures and tables excluded) containing:

- Your answers to Tasks 2 and 3
- A description of the agents of Task 4
- Your results and discussion for Task 5
- Task 6, if done

Contributions: Please include a section that describes the work each team member contributed to the project. This ensures team members carry a fair share of the work. This section does not count towards the total word count.

Deliverables:

One submission per team on Moodle, before 13 Nov 2026 at 23:59. Upload a zip file containing the report (PDF), agents.py, experiment.ipynb, the JSON log of each team member, and any instructions required to run your code.

Important:

- Failing to submit a report leads to a mark of zero (0).
- Failing to run your code leads to a mark of zero (0).
- If an AI Assistant is used, failing to report it and explain its use leads to a mark of zero (0).
- A team member who did not play the game loses 2 points.
- Teams drawn at random will have a 10-minute oral in the week of 16 Nov 2026. Any member may be asked any question. The oral can change the team or individual grade.